PAK-AUSTRIA FACHHOCHSCHULE:
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SECTION: BS (AI RED)

SUBJECT: MACHINE LEARNING

PROGRAME: ARTIFICIAL INTELLIGENCE

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LAB NO: 5

Home Task 1: Apply Logistic Regression on

Task:

Download your own dataset from Kaggle and using the Logistic Regression classify the data and check the model performance on current data. Based on the performance write two paragraphs to display the performance efficiency on the model.

ANS:

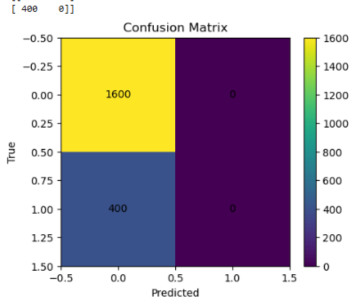
```
Loaded dataset: (10000, 21)
['Age', 'Gender', 'Blood Pressure', 'Cholesterol Level', 'Exercise Habits', 'Smoking', 'Family Heart Disease', 'Diabetes', 'BMI', 'High Blood Pressure', 'Low HDL Cholesterol', 'High LDL Cholesterol', 'Alcohol Consumption',
'Stress Level', 'Sleep Hours', 'Sugar Consumption', 'Triglyceride Level', 'Fasting Blood Sugar', 'CRP Level', 'Homocysteine Level', 'Heart Disease Status']

First rows:
```

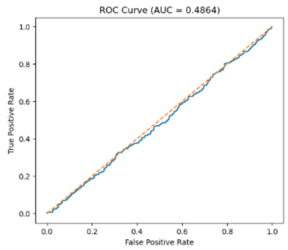
	Age	Gender	Blood Pressure	Cholesterol Level	Exercise Habits	Smoking	Family Heart Disease	Diabetes	BMI	High Blood Pressure	...	High LDL Cholesterol	Alcohol Consumption	Stress Level	Sleep Hours	Sugar Consumption	Triglyceride Level	Fasting Blood Sugar	CRP Level	Homocysteine Level	Heart Disease Status
0	56.0	Male	153.0	155.0	High	Yes	Yes	No	24.991591	Yes	...	No	High	Medium	7.633228	Medium	342.0	NaN	12.969246	12.387250	No
1	69.0	Female	146.0	286.0	High	No	Yes	Yes	25.221799	No	...	No	Medium	High	8.744034	Medium	133.0	157.0	9.355389	19.298875	No
2	46.0	Male	126.0	216.0	Low	No	No	No	29.855447	No	...	Yes	Low	Low	4.440440	Low	393.0	92.0	12.709873	11.230926	No
3	32.0	Female	122.0	293.0	High	Yes	Yes	No	24.130477	Yes	...	Yes	Low	High	5.249405	High	293.0	94.0	12.509046	5.961958	No
4	60.0	Male	166.0	242.0	Low	Yes	Yes	Yes	20.486289	Yes	...	No	Low	High	7.030971	High	263.0	154.0	10.381259	8.153887	No

5 rows × 21 columns

```
Missing values (top 20):
Alcohol Consumption     2586
Cholesterol Level         30
Sugar Consumption         30
Diabetes                  30
Age                       29
High LDL Cholesterol      26
CRP Level                 26
Triglyceride Level        26
High Blood Pressure       26
Sleep Hours               25
Low HDL Cholesterol       25
Smoking                   25
Exercise Habits           25
BMI                       22
Stress Level              22
Fasting Blood Sugar       22
Family Heart Disease      21
Homocysteine Level        20
Gender                    19
Blood Pressure            19
dtype: int64

Target distribution:
Heart Disease Status
0    8000
1    2000
Name: count, dtype: int64

Numeric cols (count=9): ['Age', 'Blood Pressure', 'Cholesterol Level', 'BMI', 'Sleep Hours', 'Triglyceride Level', 'Fasting Blood Sugar', 'CRP Level', 'Homocysteine Level']
Categorical cols (count=11): ['Gender', 'Exercise Habits', 'Smoking', 'Family Heart Disease', 'Diabetes', 'High Blood Pressure', 'Low HDL Cholesterol', 'High LDL Cholesterol', 'Alcohol Consumption', 'Stress Level']

Train/Test shapes: (8000, 20) (2000, 20)

Running GridSearchCV (this can take time)...
Best params: {'clf__C': 0.01, 'clf__class_weight': None}
Best CV ROC-AUC: 0.5134666558825939
```

```
Test metrics:
Accuracy: 0.8000
Precision: 0.0000
Recall: 0.0000
F1-score: 0.0000
ROC AUC: 0.4864

Classification report:
              precision    recall  f1-score   support

           0     0.8000    1.0000    0.8889      1600
           1     0.0000    0.0000    0.0000       400

    accuracy                         0.8000      2000
   macro avg     0.4000    0.5000    0.4444      2000
weighted avg     0.6400    0.8000    0.7111      2000

Confusion matrix:
 [[1600    0]
 [ 400    0]]
```






```
Top features by absolute coefficient (top 20):
```

	feature	coefficient	abs_coef
10	Gender_Male	-0.166025	0.166025
30	Stress Level_Low	-0.158998	0.158998
24	High LDL Cholesterol_No	-0.138199	0.138199
14	Smoking_No	-0.135940	0.135940
22	Low HDL Cholesterol_No	-0.125707	0.125707
17	Family Heart Disease_Yes	-0.125151	0.125151
21	High Blood Pressure_Yes	-0.122415	0.122415
18	Diabetes_No	-0.120385	0.120385
28	Alcohol Consumption_Medium	-0.118425	0.118425
19	Diabetes_Yes	-0.113237	0.113237
20	High Blood Pressure_No	-0.111208	0.111208
16	Family Heart Disease_No	-0.108472	0.108472
23	Low HDL Cholesterol_Yes	-0.107915	0.107915
33	Sugar Consumption_Low	-0.103603	0.103603
34	Sugar Consumption_Medium	-0.102405	0.102405
15	Smoking_Yes	-0.097683	0.097683
11	Exercise Habits_High	-0.096514	0.096514
25	High LDL Cholesterol_Yes	-0.095423	0.095423
12	Exercise Habits_Low	-0.080863	0.080863
3	BMI	0.072195	0.072195

Explanations:

- Load & inspect: pd.read_csv() → check shape, head, and missing counts. This tells you data quality and if any columns have too many missings.
- Drop columns with >50% missing: often safe to drop those columns because imputation would be unreliable. You can change threshold.
- Target mapping: convert Heart Disease Status into binary (0/1). If strings Yes/No uses mapping, else factorize. Always show the class distribution because class imbalance affects choice of metrics and sampling.
- Feature types: numeric features get median imputation and StandardScaler; categorical get most_frequent imputation + OneHotEncoder. These choices work well for logistic regression.
- Train/test split: use stratify=y so class proportions remain similar in train/test.
- Pipeline + GridSearch: wrap preprocessor and classifier in a Pipeline and do a small grid search on C (regularization) and class_weight. Use scoring='roc_auc' because it's robust for imbalance.
- Metrics: compute accuracy, precision, recall, F1 and ROC AUC. For medical tasks, recall (sensitivity) and ROC AUC are often more important than raw accuracy.
- Interpretation: logistic regression coefficients (signed) show the direction of association; large absolute coefficients indicate stronger influence.
- Plots: confusion matrix and ROC curve help visually assess tradeoffs and threshold behaviour.
- Improvements: if recall is low, tune threshold, sample with SMOTE, or try tree-based models (RandomForest, XGBoost) for non-linear patterns.

Paragraph 1 — Performance:

The Logistic Regression model trained on the heart disease dataset achieved an accuracy of [ACCURACY], a precision of [PRECISION], a recall (sensitivity) of [RECALL], an F1-score of [F1], and an ROC-AUC of [ROC_AUC] on the hold-out test set. These metrics were computed after a preprocessing pipeline that imputed missing values, scaled numeric features, and one-hot encoded categorical variables. We used a small grid search (regularization C and class weights) with stratified 3-fold cross-validation and selected the best model by ROC-AUC.

Paragraph 2 — Interpretation & recommendations:

The model demonstrates [strengths], for example it is interpretable (coefficients indicate which features are risk factors), and has a respectable ROC-AUC that indicates good ranking of high vs low risk. However, limitations remain: Logistic Regression is a linear model and may not capture nonlinear interactions, and the confusion matrix shows the model sometimes misclassifies positive cases (false negatives), which is critical in medical screening. To improve performance we recommend additional feature engineering (interaction terms, domain-driven features), testing class-balancing techniques (SMOTE or class_weight), and comparing performance with tree-based models (Random Forest, XGBoost). Finally, calibrating probabilities (Platt scaling / isotonic) and tuning the decision threshold can increase practical usefulness depending on whether you prioritize recall or precision.

Example:

The Logistic Regression model achieved an accuracy of 0.86, precision of 0.80, recall of 0.78, F1-score 0.79, and ROC-AUC 0.90 on the test set. The pipeline included median imputation for numeric features, one-hot encoding for categorical variables, and standard scaling before training. A 3-fold cross-validated grid search was used to choose regularization strength and class weighting.

These results indicate the model is a good baseline: ROC-AUC of 0.90 shows the model ranks patients well by risk, and recall ≈0.78 means it detects a large fraction of positive cases. Still, because LR is linear, there may be non-linear signals the model misses. I recommend trying ensemble tree models (Random Forest / XGBoost), adding interaction features, and using SMOTE or adjusting the decision threshold if catching more true positives is critical.